Chapter 4.3–4.5: Relations and Functions in Formal Methods

- Based strictly on O’Regan Ch 4.3–4.5
- Binary Relations
- Functions
- Applications in Formal Specification

4.3 Binary Relations – Formal Definition

- Given sets A and B, a binary relation $R \subseteq A \times B$
- $A \times B = \{(a, b) \mid a \in A \land b \in B\}$
- $(a, b) \in R$ means a is related to b
- Relations are sets of ordered pairs

Domain and Range of a Relation

- Domain(R) = $\{a \in A \mid \exists b: (a, b) \in R\}$
- Range(R) = $\{b \in B \mid \exists a: (a, b) \in R\}$
- Important for defining system inputs/outputs

Inverse and Composition of Relations

- Inverse: $R■^{1} = \{(b, a) \mid (a, b) \in R\}$
- Composition: $R ■ S = \{(a, c) \mid \exists b: (a, b) \in S \land (b, c) \in R\}$
- Used in state transition modeling

Properties of Relations

- Reflexive: $(a, a) \in R$ for all a
- Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$
- Transitive: $(a, b) \wedge (b, c) \Rightarrow (a, c)$
- Antisymmetric: $(a, b) \wedge (b, a) \Rightarrow a = b$

Equivalence Relations

- Reflexive + Symmetric + Transitive
- Induces equivalence classes
- Partitions a set into disjoint subsets
- Used in abstraction and state grouping

Partial Orders

- Reflexive + Antisymmetric + Transitive
- Models hierarchy and dependency
- Examples: $\leq$ on numbers, subset $\subseteq$
- Foundation for ordering in specifications

Applications of Relations (Ch 4.3.4)

- Modeling state transitions
- Representing accessibility between states
- Formal definition of system constraints
- Used in verification reasoning

4.4 Functions – Formal Definition

- A function $f \subseteq A \times B$
- For every $a \in A$, exactly one $b \in B$ exists
- Notation: $f: A \rightarrow B$
- Functions are deterministic relations

Total vs Partial Functions

- Total function: defined for every element of A
- Partial function: defined for some elements only
- Partiality important in formal specs (undefined states)

Injective, Surjective, Bijective

- Injective: $f(a1)=f(a2) \Rightarrow a1=a2$
- Surjective: $\forall b \in B, \exists a \in A: f(a)=b$
- Bijective: both injective and surjective
- Bijective $\Rightarrow$ invertible

Function Override and Update (Specification Use)

- $f \oplus \{(a, b)\}$ replaces mapping of a
- Common in state update modeling
- Used in formal system transitions

4.5 Applications of Functions

- Formal specification of program behavior
- Mapping input states to output states
- Functional programming foundations
- Supports mathematical reasoning about correctness

Functions in State-Based Modeling

- State represented as function of variables
- Transition defined as function on state
- NextState = f(CurrentState)

Relations vs Functions (Conceptual Distinction)

- All functions are relations
- Not all relations are functions
- Functions require uniqueness of output
- Relations may allow non-determinism

Relevance to Formal Methods

- Precise mathematical modeling
- Foundation for Z, VDM, and B-Method
- Used in refinement and verification
- Enables proof-based correctness

Chapter 4.3–4.5 Key Takeaways

- Relations model structured connections
- Properties determine behavioral guarantees
- Functions enforce determinism
- Essential for formal specification languages